

Reckoning on Recoveries

Radical Raters

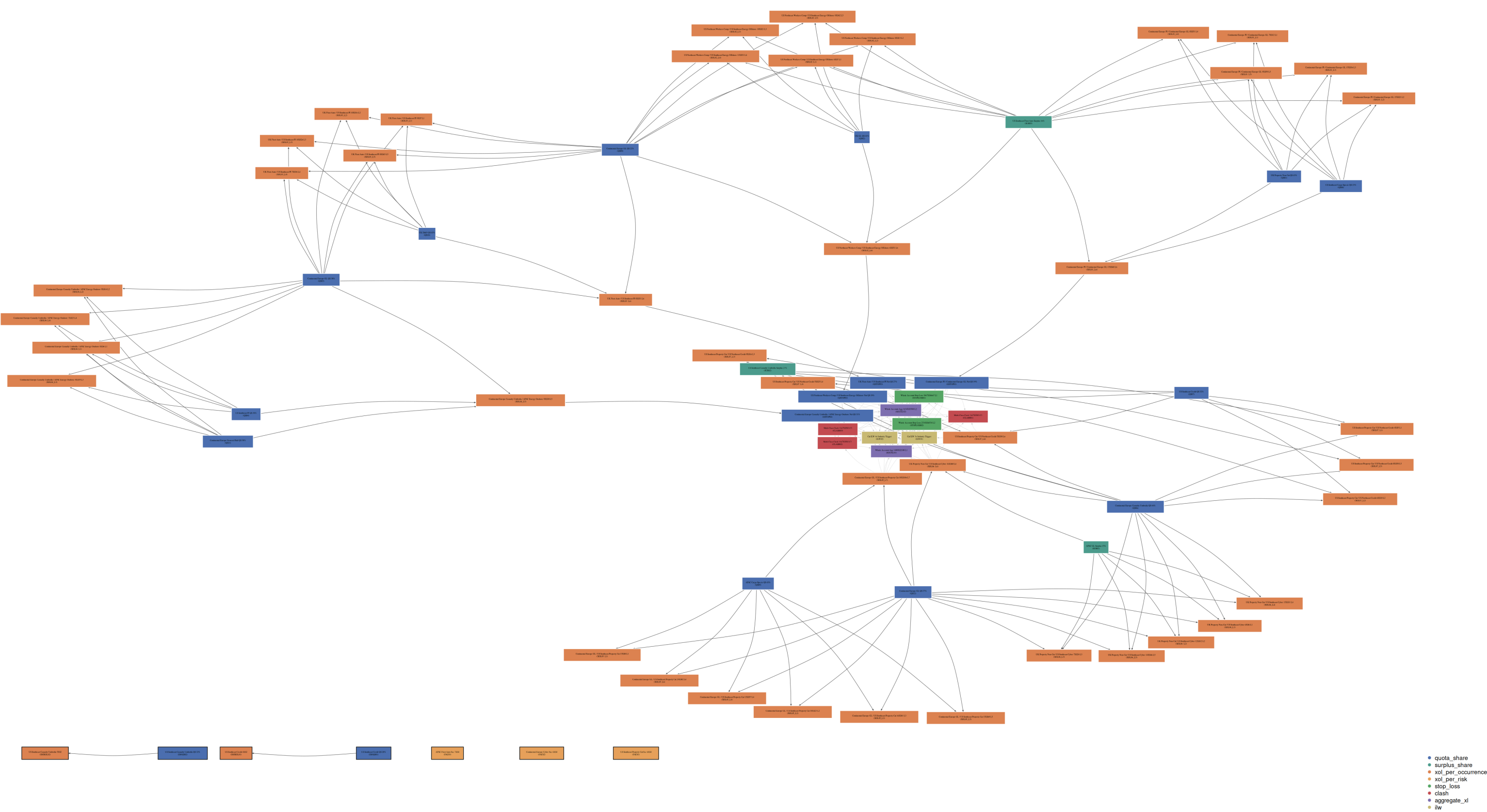
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The Problem

event_id	description	gross_loss
EV01	Small attritional loss (Tower 1 LOB)	\$18,668,971
EV02	Large loss, exhausts Tower 1 lower layers	\$233,362,133
EV03	Large loss on Tower 3 LOB (through sandwich)	\$210,025,919
EV04	Multi-LOB catastrophe -> clash hub	\$1,283,491,730
EV05	Mega-cat -> exhausts clash, into agg/ILW/stop loss	\$4,200,518,389
EV06	Small attritional loss (Tower 2 LOB)	\$14,001,728

Who pays what?

Treaty Inuring DAG — 77 treaties, 210 edges, 6 components (7 standalone outside the main structure)



How to allocate losses across treaties?

Treaties as Functions

Treaty : Losses $\rightarrow$ Recoveries

Quota Share (QS)

Excess of Loss (XoL)

Aggregate / Occurrence Limits

Recovery Accounting

10XS10 XoL

Segment	Gross Loss
Construction	5.0M
Legal	10.0M
I.T.	5.0M
Total	20.0M

Treaty recovers 10M

How do we allocate them?

Segment	Treaty	Recovery
Construction	10XS10	2.5M
Legal	10XS10	5.0M
I.T.	10XS10	2.5M
Total	10XS10	10.0M

25% QS

Segment	Gross Loss
Construction	5.0M
Legal	10.0M
I.T.	5.0M
Total	20.0M

Treaty recovers 5M

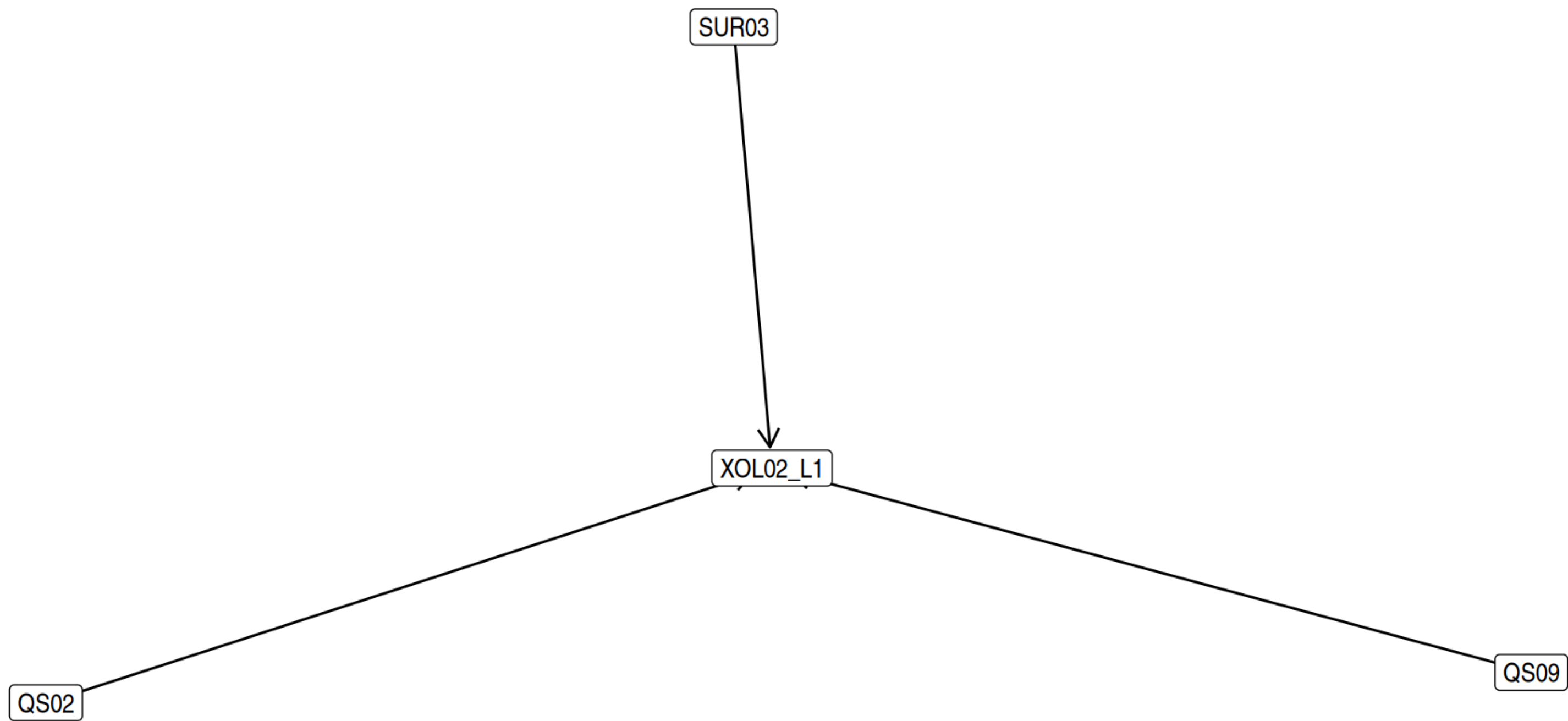
Segment	Treaty	Recovery
Construction	25% QS	1.25M
Legal	25% QS	2.50M
I.T.	25% QS	1.25M
Total	25% QS	5.00M

Inuring

Sequencing of treaties

Gross / Subject Losses

$$\text{Subject Loss} = \text{Gross Loss} - \text{Inuring Recoveries}$$



Topological Sort

Inuring graph dictates calculation order



rank	treaty_id	treaty_type	treaty_label	description
1	QS12	quota_share	Continental Europe GL QS 37%	QS on lob_5ec9f2cf609e0409, 37% ceded
2	QS03	quota_share	Continental Europe GL QS 38%	QS on lob_072509d90dee0973, 38% ceded
3	QS05	quota_share	APAC Cargo Specie QS 43%	QS on lob_7dbc7e4c7c380b86, 43% ceded
4	SUR01	surplus_share	APAC GL Surplus 15%	Surplus Share on lob_5c340bb16d8b1403, 15% ceded
5	SUR02	surplus_share	US Southeast Casualty Umbrella Surplus 27%	Surplus Share on lob_48e5e1a412975a55, 27% ceded
6	QS10	quota_share	UK D&O QS 43%	QS on lob_144138733b9281ac, 43% ceded
7	QS04	quota_share	US Southeast PI QS 38%	QS on lob_6fd96135091380a0, 38% ceded
8	QS09	quota_share	Continental Europe GL QS 22%	QS on lob_5a151f644f09e803, 22% ceded
9	QS07	quota_share	US Southeast Credit QS 32%	QS on lob_31ba4be968fa1714, 32% ceded
10	QS02	quota_share	UK GL QS 41%	QS on lob_60eae97305b9d0bc, 41% ceded
11	QS11	quota_share	Continental Europe General Hull QS 38%	QS on lob_2bf3407d167ed9d1, 38% ceded
12	SUR03	surplus_share	US Southeast Fleet Auto Surplus 16%	Surplus Share on lob_0704378d4b16e6b1, 16% ceded
13	QS01	quota_share	UK Property Non-Cat QS 43%	QS on lob_7945c1570c9592db, 43% ceded
14	QS06	quota_share	Continental Europe Casualty Umbrella QS 44%	QS on lob_6d41f7d04ca5523b, 44% ceded
15	QS08	quota_share	US Southeast Cargo Specie QS 23%	QS on lob_65066c667f902fa2, 23% ceded

Orchestration

1. Construct graph
2. Construct treaty functions
3. Topological sort
4. Loop over each node:
 - a. List inuring treaties
 - b. List inuring losses/recoveries
 - c. Keep covered losses/recoveries
 - d. Calculate subject losses
 - e. Calculate recoveries
5. Gather all per-node outputs
6. Construct outputs

Treaty / Loss coverage non-trivial

Multiple ways to determine coverage

Many-to-many relationship

Gross / net premium losses

Organisation unit

Regulatory authority

Wrap-Up

This talk...

Slides and code...